

UNICEF Education Performance Dashboard Report

Kano State, Nigeria

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Date: **May 2026**

Executive Summary

This report presents an analysis of simulated education performance indicators across selected schools in Kano State, Nigeria. The project was developed to demonstrate how data analytics tools can support planning, monitoring, and decision-making in the education sector.

The analysis focused on school enrollment, attendance rates, teacher availability, gender participation, and academic performance. Findings indicate differences in education outcomes across Local Government Areas (LGAs), with stronger performance in some urban locations and lower indicators in selected rural communities.

The dashboard provides a practical tool for identifying service gaps, improving resource allocation, and strengthening evidence-based education interventions.

1. Introduction

Education remains one of the most important drivers of social and economic development. Access to quality education improves literacy, employment opportunities, and long-term community wellbeing.

To improve educational outcomes, stakeholders require timely and accurate data to monitor school performance, identify underserved areas, and allocate resources efficiently.

This project was designed to simulate how an interactive dashboard can be used by development partners, government agencies, and program managers to track education indicators in Kano State.

2. Project Objectives

The main objectives of this project were:

- Analyze school enrollment levels across LGAs
- Measure attendance performance among students
- Assess teacher availability and workload
- Compare academic outcomes across schools
- Review gender participation trends
- Build an interactive dashboard for decision-makers

3. Data Source and Methodology

The dataset used in this project was programmatically generated using Python to simulate realistic school-level education indicators for Kano State.

Variables included:

- School Name
- LGA
- Male Enrollment
- Female Enrollment
- Total Enrollment
- Attendance Rate
- Number of Teachers
- Student-Teacher Ratio
- Exam Pass Rate

The generated dataset was cleaned and organized using Microsoft Excel. Summary statistics and pivot analysis were performed before importing the final dataset into Power BI for dashboard development.

4. Key Performance Indicators (KPIs)

The dashboard tracks the following KPIs:

- Total Number of Schools
- Total Student Enrollment
- Male Enrollment
- Female Enrollment
- Average Attendance Rate
- Number of Teachers
- Student-Teacher Ratio
- Average Pass Rate
- Best Performing LGA
- Lowest Attendance LGA

5. Key Findings

Based on the analysis, the following findings were observed:

5.1 Enrollment Trends

Urban LGAs recorded higher student enrollment compared to some rural LGAs. This may reflect better access to schools, population density, and stronger infrastructure.

5.2 Attendance Performance

Attendance levels varied across locations. Some schools recorded consistently strong attendance, while others showed weaker participation.

5.3 Teacher Distribution

Several schools had higher student-teacher ratios than recommended levels, suggesting a need for improved teacher deployment.

5.4 Gender Participation

Although male and female enrollment levels were relatively balanced overall, selected areas showed lower female participation.

5.5 Academic Performance

Exam pass rates differed across schools and LGAs, indicating variation in learning outcomes and school effectiveness.

6. Dashboard Insights

The Power BI dashboard enables users to:

- Filter results by LGA
- Compare enrollment across schools
- Review gender distribution
- Monitor attendance trends
- Identify schools with teacher shortages
- Compare exam performance across locations

The dashboard improves visibility and allows faster decision-making compared to static spreadsheet reports.

7. Recommendations

Based on the analysis, the following actions are recommended:

1. Increase teacher deployment to schools with high student-teacher ratios.
2. Strengthen attendance monitoring systems in low-performing schools.
3. Expand girl-child education campaigns in areas with lower female enrollment.
4. Prioritize interventions in LGAs with weaker pass rates.
5. Use monthly dashboard reviews for continuous performance monitoring.
6. Improve school data collection systems for real-time reporting.

8. Limitations

This project used a simulated dataset created for learning and portfolio demonstration purposes. It does not represent official administrative records.

However, the data structure and values were designed to reflect realistic education monitoring scenarios.

9. Conclusion

This project demonstrates how education data can be transformed into clear, interactive, and decision-ready insights using Python, Excel, and Power BI.

Modern dashboard systems can help education stakeholders improve planning, transparency, accountability, and service delivery.

The project highlights the practical value of analytics in supporting education development outcomes.

10. Tools Used

- Python (Dataset Generation)
- Microsoft Excel (Cleaning & Analysis)
- Power BI (Dashboard Development)
- PowerPoint / Word (Presentation & Reporting)

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